Exploring Novel Methods for Modulating Tumor Blood Vessels in Cancer Treatment

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Several studies have explored the potential of targeting tumor angiogenesis in cancer treatment. Anti-angiogenesis monotherapy, which reduces blood vessel numbers, may still hold some promise in cancer treatment, but thus far it has only provided a modest effect on overall survival benefits. When combined with standard chemotherapies, some significant improvements in cancer therapy have been reported. However, anti-angiogenesis therapies can have undesirable effects, including the induction of tumor hypoxia and reduction of delivery of chemotherapeutic drugs. Interestingly, anti-angiogenic drugs, such as bevacizumab, when used at lower doses, can actually induce vascular normalization (that is, they improve blood vessel function and flow) and potentially enhance co-administrated chemotherapeutic drug delivery. Unfortunately, vascular normalization is a difficult approach to apply in clinical settings. Thus, there is an urgent need to explore new approaches for modulating the tumor vasculature. Here, we explore how vascular promotion strategies (which enhance blood vessel numbers and leakiness) may be optimized for combination therapies as an alternative option for cancer treatment.

Introduction

Tumor angiogenesis involves the formation of new blood vessels from pre-existing blood vessels [1]. When solid tumors reach around 2 mm in diameter, they can no longer obtain sufficient nutrients and oxygen simply by diffusion from their surrounding environments. To trigger tumor angiogenesis, tumor cells secrete high levels of several important pro-angiogenic factors, including vascular endothelial growth factor-A (VEGF-A), placental growth factor (PIGF) and basic fibroblast growth factor (bFGF), into the tumor microenvironment [2]. These factors collectively activate quiescent endothelial cells of the surrounding blood vessels, which in turn promote endothelial cell proliferation, migration and tube formation via the regulation of cell adhesion molecules, such as integrins [3]. As angiogenesis is an important rate-limiting step in tumor growth, anti-angiogenesis therapy was conceived as a golden opportunity for cancer treatment [4]. It is clear that, in combination with chemotherapies, anti-angiogenic drugs (which reduce blood vessel numbers in the tumor) can provide advantages, although anti-angiogenic monotherapy is less effective [5]. For example, treatment of advanced colorectal cancer with anti-angiogenesis drugs in combination with chemotherapy can have significant survival benefits [6]. In contrast to these promising results, anti-angiogenic agents in combination with chemotherapy have consistently failed to improve overall survival for other types of cancer, including breast, pancreatic and prostate [7]. The lack of anti-angiogenic monotherapy efficacy may be due to the development of intrinsic or acquired resistance to therapy [8]. Moreover, although not proven clinically, preclinical data indicate that decreasing the tumor's blood supply results in hypoxia, possibly fueling tumor progression and metastasis [9,10].

As an alternative to anti-angiogenesis therapy, a different approach to improve blood vessel functionality has been suggested, namely vascular normalization. This strategy often involves partially reducing blood vessel numbers in tumors, whilst allowing better-functioning vessels with improved pericyte coverage and a more intact basement membrane. These vessels are less tortuous and allow for improved blood flow and reduced tumor hypoxia. Thus, in combination with chemotherapy, such vascular normalization strategies can improve drug delivery and cancer growth control [11,12]. For example, when the anti-angiogenesis drug bevacizumab is used at lower doses, it can transiently 'normalize' tumor blood vessels and partially reduce blood vessel density, resulting in an increase in tumor perfusion and providing a therapeutic window for better chemotherapeutic drug delivery [12]. Many anti-angiogenic drugs such as sorafenib and sunitinib have also been shown to act as potential normalizing agents in both preclinical and clinical models, with the functionality of tumor blood vessels being significantly improved in some cases [13,14]. Overall, however, this approach is dose and context dependent. For instance, the dose of anti-VEGF drugs needs to be adjusted relative to the level of VEGF in tumors during treatment, over time, in order to achieve an optimal balance between pro-angiogenic and anti-angiogenic factors for vessel normalization to occur. Like anti-angiogenic monotherapy, it is challenging to identify the agents and biomarkers that can be used in the clinic to provide and predict precise vessel normalization effects [15]. For comprehensive reviews of anti-angiogenesis and blood vessel normalization, please see Jain [11] and Kerbel [16].

In this review, we will discuss the current progress of anti-angiogenesis and vascular normalization approaches, and how we have learnt from them to develop an alternative therapeutic strategy for targeting the tumor vasculature that involves vascular promotion.

Anti-angiogenesis Therapy — Hopes, Victories and Limitations

Although many pro-angiogenic factors have been identified as potential targets of anti-angiogenic agents, the VEGF signaling pathway plays a critical role in stimulating angiogenesis and has been studied most extensively [17]. VEGF signaling in endothelial cells is mediated by VEGF receptor (VEGFR) tyrosine kinases and the neuropilin co-receptors [18]. The key pro-angiogenic activities of VEGF include endothelial cell proliferation, migration, sprouting and tube formation [17]. Given its major role in angiogenesis, the most successful anti-angiogenic approaches to date are those that inhibit the VEGF signaling pathway, such as bevacizumab [19]. In combination with chemotherapy or radiotherapy, treatment with bevacizumab provides promising therapeutic efficacy in cancer patients and the potential utility of metronomic scheduling, a process whereby drugs are given at relatively low doses with breaks between administration, to further improve efficacy is evident [20]. Another class of anti-angiogenic drugs is known as small-molecule tyrosine kinase inhibitors (TKIs). Sorafenib is an FDA-approved multi-targeted TKI that has been developed to inhibit VEGFR, platelet-derived growth factor receptor-beta (PDGFR-β) and Raf1 tyrosine kinase activity [13]. This drug has provided a significant survival benefit for patients with advanced renal cell carcinoma [21] and non-resectable hepatocellular carcinoma [22]. Sunitinib is another multi-targeted TKI that inhibits VEGFR, PDFGR-β, c-Kit and RET activities in tumor angiogenesis as well as in tumor cell proliferation [14]. Sunitinib was approved by the FDA for treating advanced/metastatic renal cell carcinoma [14] and imatinib-resistant gastrointestinal stromal tumors [23]. Additionally, first-line treatment of multiple myeloma includes thalidomide, a drug with multiple modes of action, one of which is anti-angiogenic [24]. Thus, there are several lines of anti-angiogenic drugs available to us, but improvements across multiple cancer types and in new combinations are still required.

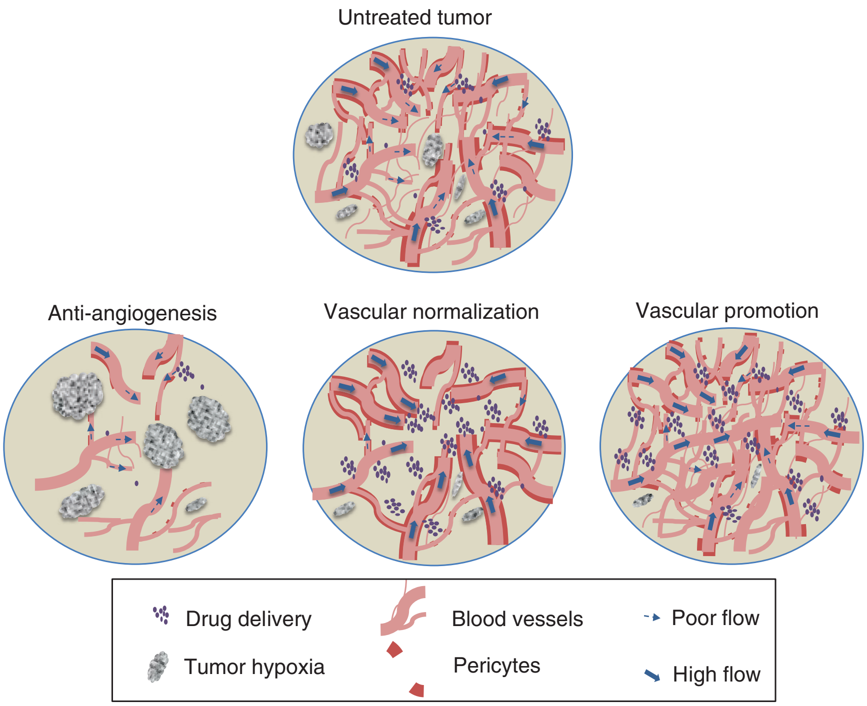
Although treatment with anti-angiogenic drugs can extend disease-free survival, the average overall survival is not improved significantly [8]. The disappointing responses may be at least in part due to a process called evasive resistance, an adaptation that overcomes the specific angiogenic blockade. Indeed, resistance to VEGF inhibition can be controlled by upregulation or modulation of other angiogenic factors such as bFGF, angiopoietins and transforming growth factor-β (TGF-β) [8]. Recently, targeting endothelial cell metabolism as a way to inhibit angiogenesis has been extensively studied and may provide a new route for cancer therapies [25]. The angiogenic switch has traditionally been thought to be dictated by angiogenic growth factors, but some findings show that it is also driven by metabolic changes within the endothelial cells. Furthermore, these changes in metabolism may override signals that induce vessel sprouting [26]. In line with this, a recent report has proposed that targeting endothelial cell metabolism could be a way to circumvent evasive resistance [27]. More specifically, genetic and pharmacological inhibition of the glycolytic regulator PFKFB3 leads to reduced angiogenesis and impairs vascular hyperbranching [28]. These data provide a new and interesting angle to control tumor angiogenesis.

In addition to evasive resistance, the majority of the available anti-angiogenic drugs cause side effects, such as severe spontaneous internal bleeding. Moreover, preclinical studies suggest that anti-angiogenic strategies elevate intratumoral hypoxia and result in increased tumor metastasis and chemotherapeutic drug resistance [9,10]. Impairing blood supply to tumors can significantly inhibit their uptake of co-administrated chemotherapeutic agents and also reduces oxygen levels within the tumors, thus inhibiting the efficacy of radiation therapy [11] (Figure 1). Thus, the future success of anti-angiogenic therapy still requires some refinement in order for it to be a broad approach for cancer treatment.

Normalization of the Tumor Vasculature by Anti-angiogenic Agents Enhances Chemotherapeutic Drug Delivery

Previous studies indicate that bevacizumab in combination with chemotherapy improves the progression-free survival of metastatic colorectal cancer patients [12]. Investigations by Jain *et al.* [29] reported that bevacizumab not only reduces tumor blood vessel density but can also temporarily normalize abnormal tumor vasculature, resulting in improved blood perfusion and drug delivery. In preclinical studies, vascular normalization was also shown to limit metastatic dissemination due to enhanced endothelial cell barrier functions and a reduction in tumor hypoxia. In contrast, studies by Van der Valdt *et al.* [30] suggested that vessel normalization was not detected in patients treated with bevacizumab, perhaps indicating how difficult it is to control this process clinically. Mice lacking PlGF were shown to have improved tumor vessel maturation and normalization [31]. Interestingly, in this case, vascular normalization occurs without a decrease in blood vessel density, indicating that anti-angiogenesis is not a prerequisite for vascular normalization. While the molecular mechanisms underlying vascular normalization are being dissected, clinical studies have demonstrated that blood vessel normalization occurs transiently in patients. Studies also observed that the appearance of a vascular normalization phenotype usually lasts for only 1–2 days after the start of treatment, providing a narrow window for enhanced drug delivery [11,12]. Inhibition of phosphatidylinositol 3-kinase (PI 3-kinase) has also been shown to promote vascular normalization in tumor models [32]. In contrast, several other tumor vessel normalization agents, such as Sac-1004, trastuzumab, and chloroquine, may provide a relatively long-lasting effect of tumor blood vessel normalization, at least in preclinical models, providing future avenues for application of vascular normalization in the clinical setting [11,33,34].

Although some reports indicate a long-lasting effect of vascular normalization, others demonstrate the more temporary nature of some vascular normalizing strategies. This may be due to the continuous high-dose anti-angiogenic treatment, which in turn favors destruction of the tumor vasculature to an extent that eventually inhibits drug delivery. Another possibility is that tumors become resistant to the anti-angiogenic drugs and use a different pathway to recruit blood vessels. There are still some important problems to be resolved, including: can we extend this temporary therapeutic window for cancer treatment in clinical settings? Is there a more effective way to determine the working doses and treatment regimes of vessel normalization approaches for individual anti-angiogenic agents and patients? Can we overcome the drug resistance and side effects of



Vascular features	Anti-angiogenesis	Vascular normalization	Vascular promotion
Blood vessel density	↓↓ [19]	↓ [11] or nsd [31]	↑ [45]
Blood flow	↓ [49]	↑ [11]	↑ [45]
Leakiness	↓ [50]	↓ [11]	↑ [45]
Tumor hypoxia	↑↑ [2]	↓ [12]	↓ [45]
Drug delivery	↓ [30]	↑ [12]	↑ [45]
Drug resistance	↑ [9,10]	↓ [11]	↓ [45]

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Figure 1. Targeting tumor vasculature to improve cancer treatment: differences between anti-angiogenesis, vascular normalization and vascular promotion strategies.
Solid tumors typically exhibit an abnormal blood vasculature associated with poor blood flow, pericyte coverage and perfusion. Although anti-angiogenic agents inhibit tumor angiogenesis and growth, they may lead to tumor hypoxia, resulting in increased drug resistance and metastasis [9,10]. This may limit the efficacy of anti-angiogenesis agents in cancer treatment. Interestingly, some anti-angiogenic agents, under certain doses and treatment regimes, can reduce vascular abnormalities, leading to vascular normalization. This approach can enhance pericyte coverage, blood flow and the delivery of chemotherapy drugs [11]. In contrast, vascular promotion is a pro-angiogenic and potentially anti-drug resistance strategy; this is achieved by enhancing blood vessel density, leakiness (reducing pericyte coverage), blood volume, perfusion and flow, as well as by increasing oxygenation and the efficacy of the chemotherapeutic drug gemcitabine [45]. The different effects of these strategies are shown pictorially and summarized in the table below (including citations to the relevant studies), where an upward arrow signifies that the particular vascular feature is increased and a downward arrow shows that it is decreased (nsd, no significance difference). Several reviews provide examples of agents or drugs that can be used for anti-angiogenesis and blood vessel normalization: see Vasudev and Reynolds [7], Jain [11], and Carmeliet and Jain [12] for details.

anti-angiogenic agents during treatment? Answering these questions will allow a better understanding of the vascular normalization phenomenon and its potential utility in clinical settings for a variety of solid cancer types.

Vascular Promotion Can Improve Blood Supply and Chemotherapeutic Drug Delivery

Tumor blood flow, perfusion, and oxygenation are the key factors that determine sensitivity to radiotherapies and several chemotherapies. Some strategies that inhibit Delt-like ligand 4 (Dll4) or the p110α subunit of PI 3-kinase increase tumor vascularity but the resulting vessels are non-functional, leading to increased tumor hypoxia [35,36]. In contrast, some investigators have explored pro-vascular approaches that were aimed at improving blood flow to enhance oxygen supply and drug delivery [37]. One of these strategies involved using lower doses of vasodilators, such as the calcium channel antagonists diltiazem, nisoldipine, flunarizine and verapamil, all of which were shown to enhance blood flow in tumors, at least in preclinical models [38,39]. Furthermore, verapamil has also been reported to enhance chemotherapeutic drug efficacy by reversing multidrug resistance in tumors [40]. However, the maximal tolerated doses of these calcium channel antagonists used in some studies caused hypotension, limiting their use in cancer therapy. Nevertheless, newer evidence suggests that treatment with suboptimal doses of verapamil can still provide significant clinical benefits and reduce possible side effects [41].

In contrast, other studies have indicated that treatment with low doses of cilengitide, an Arg-Gly-Glu-containing pentapeptide that selectively blocks activation of the αvβ3/αvβ5 integrins, can actually increase tumor blood vessel density by enhancing endothelial cell VEGFR recycling to the cell membrane, a process called vascular promotion [42]. αvβ3/αvβ5 integrins are expressed on several cell types including endothelial cells and tumor cells. While cilengitide was originally developed as an anti-angiogenic agent, it has also been reported to significantly promote the death of $\alpha v\beta 3/\alpha v\beta 5$ integrin-positive cancer cells [43]. However, cilengitide (used at maximum doses) has recently failed in clinical trials, showing no overall benefit [44]. Thus, the opportunity to reprogram the use of this drug became available. Due to the pro-angiogenic and vasodilation features of low-dose cilengitide and verapamil, respectively, it was hypothesized that this therapeutic combination could enhance the clinical performance of chemotherapeutic agents, such as gemcitabine, whilst reducing toxic side effects. By testing this combination therapy in a range of preclinical models, including lung and pancreatic cancer, it was demonstrated that the combination of suboptimal doses of verapamil (5mg/kg/day or three times a week) and low-dose cilengitide (50μg/kg/day or three times a week) significantly increases tumor blood vessel densities, leakiness (measured by Hoechst diffusion into perivascular tissues), blood perfusion and blood supply, leading to enhanced and more efficient delivery of gemcitabine [45]. In contrast, some vascular normalizing agents are characterized by their reduction of blood vessel leakiness to improve blood vessel stability [11]. Thus, anti-angiogenic therapy, vascular normalization and vascular promotion strategies all have distinct features (Figure 1).

Briefly, vascular promotion increases blood vessel density, blood flow and blood vessel leakiness whilst reducing hypoxia [45]. Combination treatment with low-dose cilengitide, verapamil and gemcitabine reduces primary tumor growth and metastasis significantly, and more importantly extends overall survival. Unlike other approaches, this vascular promotion treatment significantly delays cancer progression, even after the termination of treatment. Interestingly, vascular promotion therapy appears to be effective in cancer types that have both poorly vascularized (i.e. pancreatic cancer), and highly vascularized (i.e. non-small cell lung cancer) tumors, suggesting its potential application across a broad range of cancers.

Low-dose Cilengitide Enhances Chemotherapeutic Efficacy

Apart from its pro-angiogenic effects, it has been shown that co-administration of low-dose cilengitide and verapamil also enhances gemcitabine metabolism and efficacy. In general, gemcitabine uptake into tumor cells is regulated by equilibrative nucleoside transporters 1 and 2 (ENT1 and ENT2), and concentrative nucleoside transporter 3 (CNT3). Indeed, treatment with low-dose cilengitide and verapamil significantly reduced hypoxia in lung and pancreatic cancer, and led to an upregulation of ENT1, ENT2 and CNT3 expression. Once inside the tumor cells, gemcitabine (dFdC) is converted into its active form (dFdCMP) by the rate-limiting enzyme deoxycytidine kinase (DCK), and then finally phosphorylated to gemcitabine triphosphate (dFdCTP) [46]. In contrast, cytidine deaminase (CDA) promotes the conversion of gemcitabine into its inactive form (dFdU) [47]. Interestingly, treatment with low-dose cilengitide enhanced DCK expression but downregulated CDA expression both *in vitro* and *in vivo*. Thus, the increase in gemcitabine uptake, combined with the decreased efflux of the drug, led to enhanced expression of DCK, which in turn generated the more active form of gemcitabine. As a result, the dual action of low-dose cilengitide and verapamil allowed for the effective use of reduced doses of gemcitabine, even when compared with its maximum tolerated dose in our preclinical models. Thus, this combination therapy significantly enhances intratumoral delivery and intracellular uptake of the cytotoxic drug and minimizes adverse effects, leading to improved chemotherapeutic drug efficacy in pre-clinical studies. These results imply that it would be worthwhile exploring the potential of vascular promotion therapy for improving the delivery and even the tumor-killing efficacy of other chemotherapeutic drugs, such as platinum-based compounds. Indeed, the reduction in hypoxia through this approach could also be useful in improving the efficacy of radiotherapy. Additional considerations of the broader application of vascular promotion therapy across several different cancer types will also be important, especially where anti-angiogenic drugs have previously failed [48].

Conclusions and Future Perspectives

Several therapeutic strategies have been developed to target tumor angiogenesis; however, none provides a completely comprehensive approach. Recent studies indicate that some anti-angiogenesis agents can functionally improve vascular perfusion in tumors, with the potential to increase chemotherapeutic drug delivery to tumor cells. However, anti-angiogenic agents can destroy tumor vasculature to an extent that induces an increase in tumor hypoxia and a decrease in drug delivery. More recent studies provide proof-of-principle that vascular promotion can significantly improve the efficacy of cancer treatment. In the future, it will be important to determine the broader potential clinical relevance of vascular promotion in the context of different chemotherapeutic drugs, other nucleoside analogs, antibody therapies, or radiotherapies.

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REFERENCES

1. Folkman, J. (2007). Angiogenesis: an organizing principle for drug discovery? Nat. Rev. Drug Discov. *6*, 273–286.
2. Chung, A.S., Lee, J., and Ferrara, N. (2010). Targeting the tumour vasculature: insights from physiological angiogenesis. Nat. Rev. Cancer *10*, 505–514.
3. Desgrosellier, J.S., and Cheresh, D.A. (2010). Integrins in cancer: biological implications and therapeutic opportunities. Nat. Rev. Cancer *10*, 9–22.
4. Folkman, J. (1971). Tumor angiogenesis: therapeutic implications. New Engl. J. Med. *285*, 1182–1186.
5. Okines, A., and Cunningham, D. (2009). Current perspective: bevacizumab in colorectal cancer–a time for reappraisal? Eur. J. Cancer *45*, 2452–2461.
6. Giantonio, B.J., Catalano, P.J., Meropol, N.J., O'Dwyer, P.J., Mitchell, E.P., Alberts, S.R., Schwartz, M.A., Benson, A.B., 3rd, Cooperative, Eastern, and Oncology Group Study, E. (2007). Bevacizumab in combination with oxaliplatin, fluorouracil, and leucovorin (FOLFOX4) for previously treated metastatic colorectal cancer: results from the Eastern Cooperative Oncology Group Study E3200. J. Clin. Oncol. *25*, 1539–1544.
7. Vasudev, N.S., and Reynolds, A.R. (2014). Anti-angiogenic therapy for cancer: current progress, unresolved questions and future directions. Angiogenesis *17*, 471–494.

8. Bergers, G., and Hanahan, D. (2008). Modes of resistance to anti-angiogenic therapy. Nat. Rev. Cancer *8*, 592–603.

9. Ebos, J.M., Lee, C.R., Cruz-Munoz, W., Bjarnason, G.A., Christensen, J.G., and Kerbel, R.S. (2009). Accelerated metastasis after short-term treatment with a potent inhibitor of tumor angiogenesis. Cancer Cell *15*, 232–239.

10. Paez-Ribes, M., Allen, E., Hudock, J., Takeda, T., Okuyama, H., Vinals, F., Inoue, M., Bergers, G., Hanahan, D., and Casanovas, O. (2009). Antiangiogenic therapy elicits malignant progression of tumors to increased local invasion and distant metastasis. Cancer Cell *15*, 220–231.

11. Jain, R.K. (2014). Antiangiogenesis strategies revisited: from starving tumors to alleviating hypoxia. Cancer Cell *26*, 605–622.

12. Carmeliet, P., and Jain, R.K. (2011). Principles and mechanisms of vessel normalization for cancer and other angiogenic diseases. Nat. Rev. Drug Discov. *10*, 417–427.

13. Wilhelm, S., Carter, C., Lynch, M., Lowinger, T., Dumas, J., Smith, R.A., Schwartz, B., Simantov, R., and Kelley, S. (2006). Discovery and development of sorafenib: a multikinase inhibitor for treating cancer. Nat. Rev. Drug Discov. *5*, 835–844.

14. Motzer, R.J., Hutson, T.E., Tomczak, P., Michaelson, M.D., Bukowski, R.M., Rixe, O., Oudard, S., Negrier, S., Szczylik, C., Kim, S.T., *et al*. (2007). Sunitinib versus interferon alfa in metastatic renal-cell carcinoma. New Engl. J. Med. *356*, 115–124.

15. Webb, T. (2005). Vascular normalization: study examines how antiangiogenesis therapies work. J. Natl. Cancer Inst. *97*, 336–337.

16. Kerbel, R.S. (2008). Tumor angiogenesis. New Engl. J. Med. *358*, 2039–2049.

17. Goel, H.L., and Mercurio, A.M. (2013). VEGF targets the tumour cell. Nat. Rev. Cancer *13*, 871–882.

18. Neufeld, G., Kessler, O., and Herzog, Y. (2002). The interaction of Neuropilin-1 and Neuropilin-2 with tyrosine-kinase receptors for VEGF. Adv. Exp. Med. Biol. *515*, 81–90.

19. Ferrara, N., Hillan, K.J., Gerber, H.P., and Novotny, W. (2004). Discovery and development of bevacizumab, an anti-VEGF antibody for treating cancer. Nat. Rev. Drug Discov. *3*, 391–400.

20. Kerbel, R.S. (2012). Strategies for improving the clinical benefit of antiangiogenic drug based therapies for breast cancer. J. Mammary Gland Biol. Neoplasia *17*, 229–239.

21. Escudier, B., Eisen, T., Stadler, W.M., Szczylik, C., Oudard, S., Siebels, M., Negrier, S., Chevreau, C., Solska, E., Desai, A.A., *et al*. (2007). Sorafenib in advanced clear-cell renal-cell carcinoma. New Engl. J. Med. *356*, 125–134.

22. Llovet, J.M., Ricci, S., Mazzaferro, V., Hilgard, P., Gane, E., Blanc, J.F., de Oliveira, A.C., Santoro, A., Raoul, J.L., Forner, A., *et al*. (2008). Sorafenib in advanced hepatocellular carcinoma. New Engl. J. Med. *359*, 378–390.

23. Demetri, G.D., van Oosterom, A.T., Garrett, C.R., Blackstein, M.E., Shah, M.H., Verweij, J., McArthur, G., Judson, I.R., Heinrich, M.C., Morgan, J.A., *et al*. (2006). Efficacy and safety of sunitinib in patients with advanced gastrointestinal stromal tumour after failure of imatinib: a randomised controlled trial. Lancet *368*, 1329–1338.

24. D'Amato, R.J., Loughnan, M.S., Flynn, E., and Folkman, J. (1994). Thalidomide is an inhibitor of angiogenesis. Proc. Natl. Acad. Sci. USA *91*, 4082–4085.

25. Vandekeere, S., Dewerchin, M., and Carmeliet, P. (2015). Angiogenesis revisited: an overlooked role of endothelial cell metabolism in vessel sprouting. Microcirculation *22*, 509–517.

26. Eelen, G., de Zeeuw, P., Simons, M., and Carmeliet, P. (2015). Endothelial cell metabolism in normal and diseased vasculature. Circ. Res. *116*, 1231–1244.

27. Verdegem, D., Moens, S., Stapor, P., and Carmeliet, P. (2014). Endothelial cell metabolism: parallels and divergences with cancer cell metabolism. Cancer Metab. *2*, 19.

28. Schoors, S., De Bock, K., Cantelmo, A.R., Georgiadou, M., Ghesquiere, B., Cauwenberghs, S., Kuchnio, A., Wong, B.W., Quaegebeur, A., Goveia, J., *et al*. (2014). Partial and transient reduction of glycolysis by PFKFB3 blockade reduces pathological angiogenesis. Cell Metab. *19*, 37–48.

29. Jain, R.K., Duda, D.G., Clark, J.W., and Loeffler, J.S. (2006). Lessons from phase III clinical trials on anti-VEGF therapy for cancer. Nature clinical practice. Oncology *3*, 24–40.

30. Van der Veldt, A.A., Lubberink, M., Bahce, I., Walraven, M., de Boer, M.P., Greuter, H.N., Hendrikse, N.H., Eriksson, J., Windhorst, A.D., Postmus, P.E., *et al*. (2012). Rapid decrease in delivery of chemotherapy to tumors after anti-VEGF therapy: implications for scheduling of anti-angiogenic drugs. Cancer Cell *21*, 82–91.

31. Rolny, C., Mazzone, M., Tugues, S., Laoui, D., Johansson, I., Coulon, C., Squadrito, M.L., Segura, I., Li, X., Knevels, E., *et al*. (2011). HRG inhibits tumor growth and metastasis by inducing macrophage polarization and vessel normalization through downregulation of PlGF. Cancer Cell *19*, 31–44.

32. Qayum, N., Im, J., Stratford, M.R., Bernhard, E.J., McKenna, W.G., and Muschel, R.J. (2012). Modulation of the tumor microvasculature by phosphoinositide-3 kinase inhibition increases doxorubicin delivery in vivo. Clin. Cancer. Res. *18*, 161–169.

33. Maes, H., Kuchnio, A., Peric, A., Moens, S., Nys, K., De Bock, K., Quaegebeur, A., Schoors, S., Georgiadou, M., Wouters, J., *et al*. (2014). Tumor vessel normalization by chloroquine independent of autophagy. Cancer Cell *26*, 190–206.

34. Agrawal, V., Maharjan, S., Kim, K., Kim, N.J., Son, J., Lee, K., Choi, H.J., Rho, S.S., Ahn, S., Won, M.H., *et al*. (2014). Direct endothelial junction restoration results in significant tumor vascular normalization and metastasis inhibition in mice. Oncotarget *5*, 2761–2777.

35. Noguera-Troise, I., Daly, C., Papadopoulos, N.J., Coetzee, S., Boland, P., Gale, N.W., Lin, H.C., Yancopoulos, G.D., and Thurston, G. (2006). Blockade of Dll4 inhibits tumour growth by promoting non-productive angiogenesis. Nature *444*, 1032–1037.

36. Soler, A., Serra, H., Pearce, W., Angulo, A., Guillermet-Guibert, J., Friedman, L.S., Vinals, F., Gerhardt, H., Casanovas, O., Graupera, M., *et al*. (2013). Inhibition of the p110alpha isoform of PI 3-kinase stimulates nonfunctional tumor angiogenesis. J. Exp. Med. *210*, 1937–1945.

37. Sonveaux, P. (2008). Provascular strategy: targeting functional adaptations of mature blood vessels in tumors to selectively influence the tumor vascular reactivity and improve cancer treatment. Radiother. Oncol. *86*, 300–313.

38. Wood, P.J., and Hirst, D.G. (1989). Modification of tumour response by calcium antagonists in the SCVII/St tumour implanted at two different sites. Int. J. Radiat. Biol. *56*, 355–367.

39. Kaelin, W.G., Jr., Shrivastav, S., Shand, D.G., and Jirtle, R.L. (1982). Effect of verapamil on malignant tissue blood flow in SMT-2A tumor-bearing rats. Cancer Res. *42*, 3944–3949.

40. Bergman, A.M., Pinedo, H.M., Talianidis, I., Veerman, G., Loves, W.J., van der Wilt, C.L., and Peters, G.J. (2003). Increased sensitivity to gemcitabine of P-glycoprotein and multidrug resistance-associated protein-overexpressing human cancer cell lines. Br. J. Cancer *88*, 1963–1970.

41. Ahmed, J.H., Elliott, H.L., Meredith, P.A., and Reid, J.L. (1992). Low-dose verapamil in middle-aged and elderly patients with angina pectoris: no evidence of increased susceptibility to the cardiac effects. Cardiovasc. Drugs Ther. *6*, 153–158.

42. Reynolds, A.R., Hart, I.R., Watson, A.R., Welti, J.C., Silva, R.G., Robinson, S.D., Da Violante, G., Gourlaouen, M., Salih, M., Jones, M.C., *et al*. (2009). Stimulation of tumor growth and angiogenesis by low concentrations of RGD-mimetic integrin inhibitors. Nat. Med. *15*, 392–400.

43. Oliveira-Ferrer, L., Hauschild, J., Fiedler, W., Bokemeyer, C., Nippgen, J., Celik, I., and Schuch, G. (2008). Cilengitide induces cellular detachment and apoptosis in endothelial and glioma cells mediated by inhibition of FAK/src/AKT pathway. J. Exp. Clin. Cancer Res. *27*, 86.

44. Stupp, R., Hegi, M.E., Gorlia, T., Erridge, S.C., Perry, J., Hong, Y.K., Aldape, K.D., Lhermitte, B., Pietsch, T., Grujicic, D., *et al*. (2014). Cilengitide combined with standard treatment for patients with newly diagnosed glioblastoma with methylated MGMT promoter (CENTRIC EORTC 26071-22072 study): a multicentre, randomised, open-label, phase 3 trial. Lancet Oncol. *15*, 1100–1108.

45. Wong, P.P., Demircioglu, F., Ghazaly, E., Alrawashdeh, W., Stratford, M.R., Scudamore, C.L., Cereser, B., Crnogorac-Jurcevic, T., McDonald, S., Elia, G., *et al*. (2015). Dual-action combination therapy enhances angiogenesis while reducing tumor growth and spread. Cancer Cell *27*, 123–137.

46. Bergman, A.M., Pinedo, H.M., and Peters, G.J. (2002). Determinants of resistance to 2',2'-difluorodeoxycytidine (gemcitabine). Drug Resis. Updates *5*, 19–33.

47. Neff, T., and Blau, C.A. (1996). Forced expression of cytidine deaminase confers resistance to cytosine arabinoside and gemcitabine. Exp. Hematol. *24*, 1340–1346.

48. Sledge, G.W. (2015). Anti-vascular endothelial growth factor therapy in breast cancer: game over? J. Clin. Oncol. *33*, 133–135.

49. Inai, T., Mancuso, M., Hashizume, H., Baffert, F., Haskell, A., Baluk, P., Hu-Lowe, D.D., Shalinsky, D.R., Thurston, G., Yancopoulos, G.D., *et al*. (2004). Inhibition of vascular endothelial growth factor (VEGF) signaling in cancer causes loss of endothelial fenestrations, regression of tumor vessels, and appearance of basement membrane ghosts. Am. J. Pathol. *165*, 35–52.

50. Pham, C.D., Roberts, T.P., van Bruggen, N., Melnyk, O., Mann, J., Ferrara, N., Cohen, R.L., and Brasch, R.C. (1998). Magnetic resonance imaging detects suppression of tumor vascular permeability after administration of antibody to vascular endothelial growth factor. Cancer Invest. *16*, 225–230.